مدارس الإمارات الوطنية
Emirates National Schools

Grade: 5 | Term 1, Week 9 (26 Oct to 30 Oct) | TCD: Future Skills 5: Prototyping and Smart Making | Session length: 1 hr 30 min

T1 WEEK: 9	**Lesson 1: Tinkercad Basics: Keychain or Name Tag**
Standard(s):	Math: 5.MD.C.3 (volume and 3D figures), 5.G.A (coordinate plane); Technology: CAD basics
Literacy Strategies:	Tinkercad tool cards with picture and word; partners explain each tool using: This tool lets me ...
Assessment:	Saved Tinkercad design checked against the size checklist; observation; live quiz on the TCD website.
WALT (Learning Target):	We are learning to use the basic Tinkercad tools to design an accurate keychain or name tag.
WILF (Success Criteria):	I can drag, resize and rotate shapes using millimetre measurements. I can use a hole shape to cut a ring hole. I can align and group shapes into one object. I can add raised text to my design.
Targeted Vocabulary	Tinkercad, workplane, millimetre, solid, hole, align, group, dimensions
Resources	Computers or iPads with Tinkercad, Tinkercad class code, Tool picture cards, Sample printed keychains and name tags, Rulers Accompanying resource: Grade5_Week9_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade5_Week9_Worksheet and Grade5_Week9_Lesson_Plan.
National Identity Integration	Students can add a simple pattern inspired by Emirati Sadu weaving or the shape of a dhow to their keychain.
AI/Digital Learning	Tinkercad accounts through the class code; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Tour the Tinkercad screen (slide 6). Model solids and holes (slide 7) and typing sizes in millimetres (slide 8). Model align, group and the text shape (slides 9 and 10).
Guided Practice: We Do	Students log in and follow along to make a practice box with a hole. Teacher checks that every student can type a size and group a hole.
Independent Practice: You Do	Students design their keychain or name tag (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Follow the step cards to build a rectangle keychain with your initials.** **On: Design a keychain with your full name and a ring hole, all sizes correct.** **Above: Add a shape or pattern (for example a dhow or a Sadu-inspired border) and round the corners.**
Closure/Reflection	Three students share their screen and explain one tool. Exit ticket (slide 15). Check every design is saved and named.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☑ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☐ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☐ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☐ Higher-Order Questioning ☐ Curriculum Compacting ☐ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☑ Open-Ended Task ☐ Other: _____